

Personal Cybersecurity Lab Build Report

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1. Objective

The objective of this task was to build a safe and isolated cybersecurity practice environment on a local computer. This hands-on lab serves as the foundation for future tasks such as vulnerability assessment, web application testing, exploitation practice, and incident response, ensuring that all testing is contained and cannot affect real-world systems.

2. Project Summary: What Was Accomplished

During this lab, a complete, self-contained penetration testing environment was built from scratch. The following key actions were performed:

- **Virtualization Setup:** Installed VirtualBox to act as the host platform for creating isolated virtual computers.
- **Attacker Environment:** Imported and configured a Kali Linux virtual machine, providing a suite of industry-standard offensive security tools.
- **Network Isolation:** Configured a dual-adapter network setup. One adapter provided safe internet access for downloading tools, while the second (Host-Only) adapter created a closed-loop network to ensure security tools and malware cannot escape into the real network.
- **Target Deployment:** Installed Docker and deployed a containerized version of OWASP Juice Shop, an intentionally vulnerable web application used for hacking practice.
- **Tool Validation:** Validated the functionality of the lab using three core cybersecurity tools: Nmap (to scan the network and discover the target), Burp Suite (to intercept and manipulate live web traffic between the browser and the target), and Wireshark (to perform live packet capture and analyze raw network data).

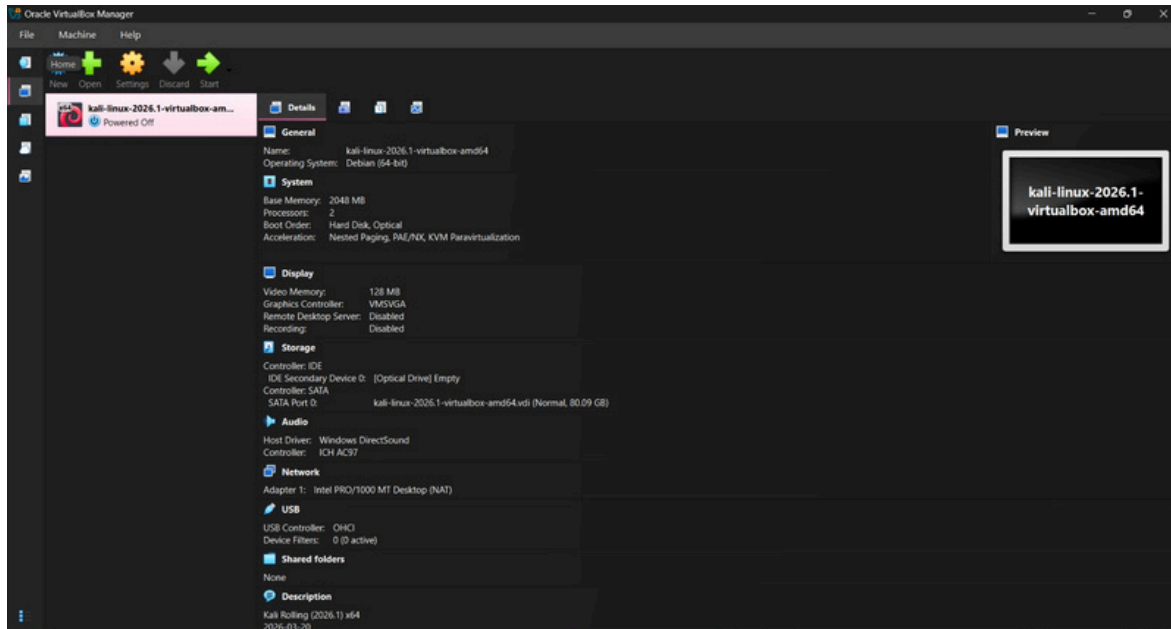
3. Architecture Overview

Host OS	Windows
Virtualization Software	Oracle VirtualBox
Attacker Machine	Kali Linux (2 CPUs, 2GB RAM)
Target Machine	OWASP Juice Shop (Deployed via Docker)
Networking – Adapter 1	NAT (Internet access)
Networking – Adapter 2	Host-Only Network (Isolated lab traffic)

4. Evidence of Implementation

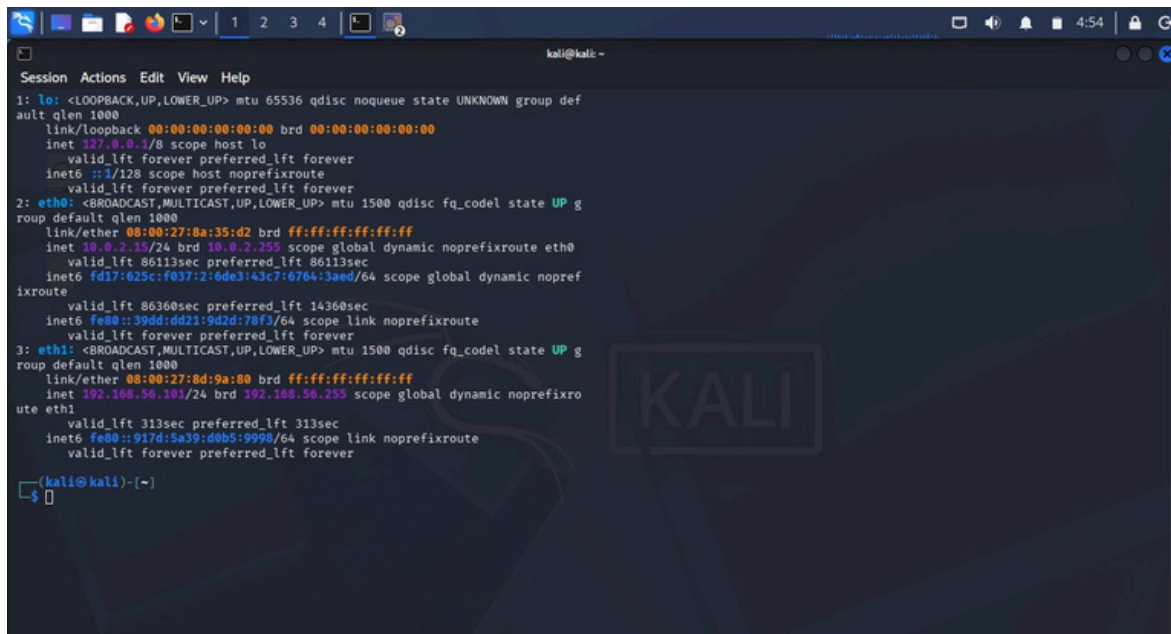
4.1 Virtual Machine Setup

The Kali Linux attacker machine was successfully imported and configured in VirtualBox.



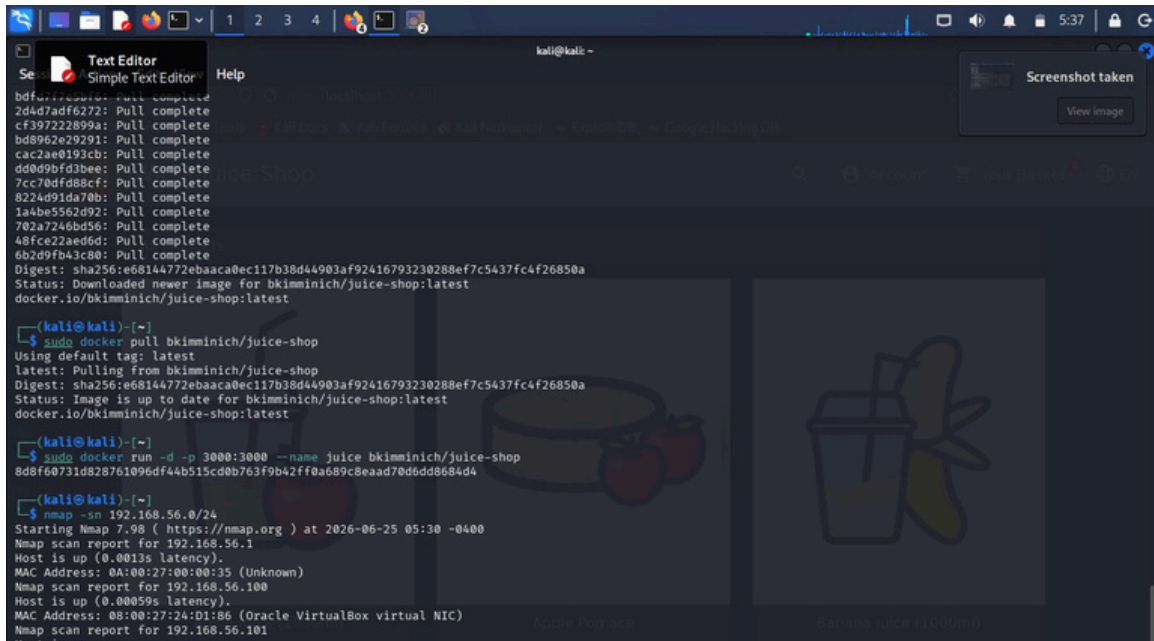
4.2 IP Configuration & Network Segmentation

The network adapters were verified inside the Kali Linux terminal using the **ip a** command, confirming both the NAT and Host-Only networks were active.



4.3 Target Deployment & Ping/Nmap Validation

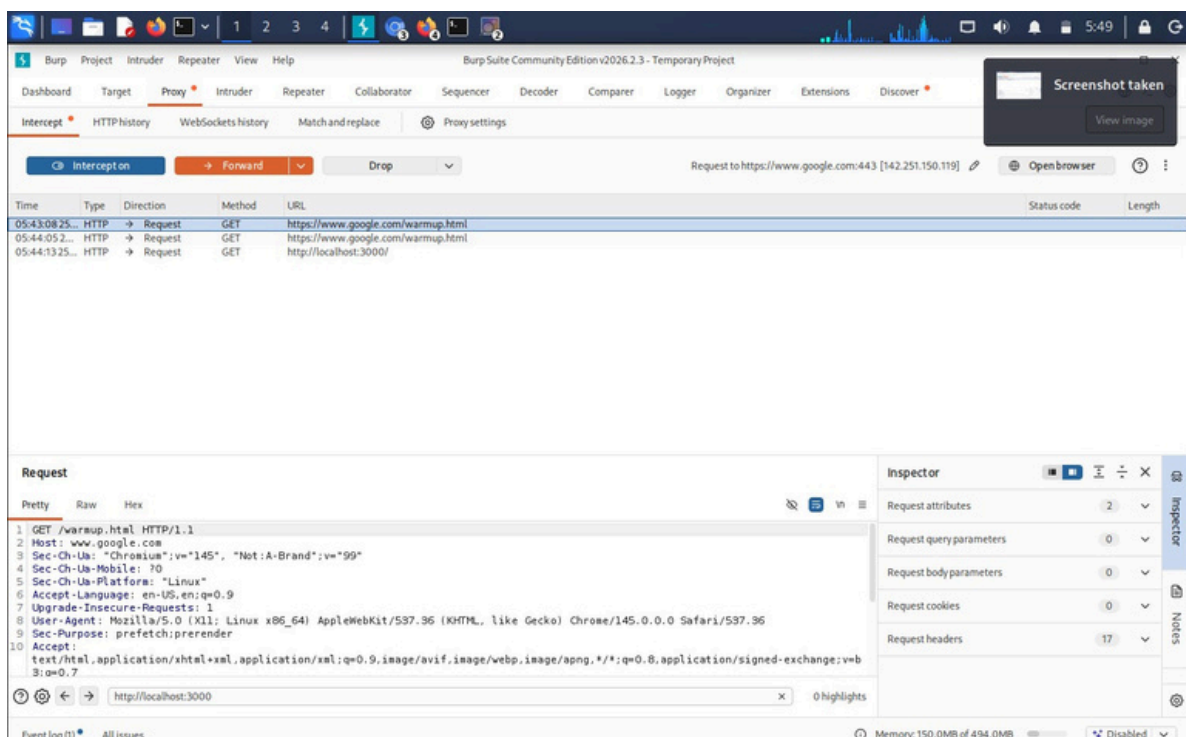
The intentionally vulnerable OWASP Juice Shop application was deployed via Docker. An Nmap scan was conducted on the 192.168.56.0/24 subnet, successfully discovering the active hosts on the isolated network.



```
kali@kali: ~  
$ docker pull bkimminich/juice-shop  
Pulling bkimminich/juice-shop  
bd8f731d828761096df44b515cd8b763f9b42ff0a689c8ead70d6dd8684d4: Pull complete  
Digest: sha256:e68144772ebaaca0ec117b38d44903af92416793230288ef7c5437fc4f26850a  
Status: Downloaded newer image for bkimminich/juice-shop:latest  
docker.io/bkimminich/juice-shop:latest  
  
$ docker run -d -p 3000:3000 --name juice bkimminich/juice-shop  
8d8f731d828761096df44b515cd8b763f9b42ff0a689c8ead70d6dd8684d4  
  
$ nmap -sn 192.168.56.0/24  
Starting Nmap 7.90 ( https://nmap.org ) at 2026-06-25 05:30 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0013s latency).  
MAC Address: 0A:00:27:00:00:35 (Unknown)  
Nmap scan report for 192.168.56.100  
Host is up (0.00059s latency).  
MAC Address: 08:00:27:24:D1:86 (Oracle VirtualBox virtual NIC)  
Nmap scan report for 192.168.56.101
```

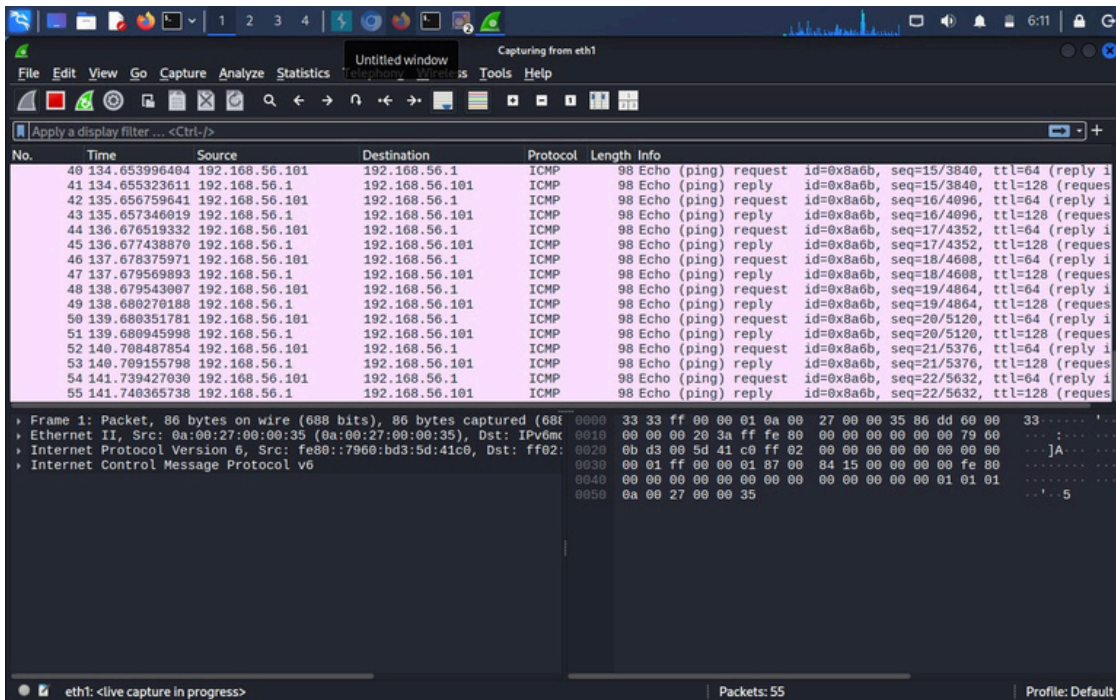
4.4 Web Application Access & Interception (Burp Suite)

Burp Suite was successfully configured as a proxy to intercept web traffic. The request to the local Juice Shop application (localhost:3000) was successfully trapped and analyzed before being forwarded to the server.



4.5 Network Traffic Analysis (Wireshark)

Wireshark was launched and attached to the eth1 (Host-Only) interface. ICMP (ping) traffic was generated to demonstrate live packet capture and network traffic analysis on the isolated lab network.



5. Conclusion

The personal cybersecurity lab has been successfully built and verified. Proper network segmentation ensures that the attacker machine and the vulnerable target are completely isolated from the host network. All required tools (Docker, Nmap, Burp Suite, and Wireshark) were successfully tested and proven to be fully operational within the lab environment.